SOFTWARE ARCHITECTURE

Client-Server architecture

BEYOND BOUNDARIES

HOA SEN UNIVERSITY
WORLD CLASS EDUCATION

QS STARS RATING SYSTEM

ACBSP ACCREDITED

AUN-QA

MOET ACCREDITED
KIỂM ĐỊNH CHẤT LƯỢNG GIÁO DỤC QUỐC GIA

Outline

- Introduction
- Synchronous vs. Asynchronous
- Model structure
- Evaluation

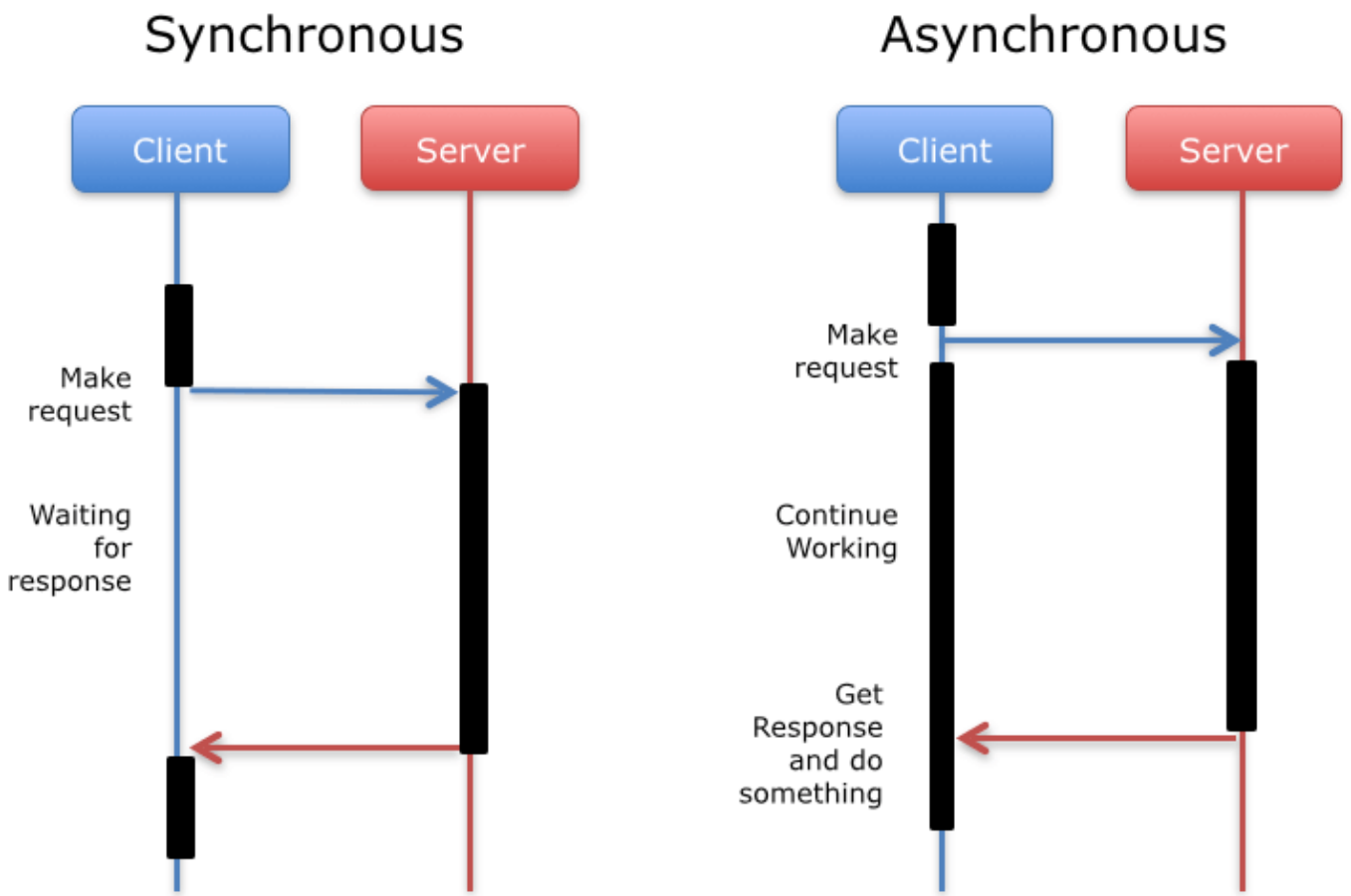
Introduction

- Client-Server is a 2-tiers architecture, in which each tier will run a standalone application to connect and exchange data with each other through the network.
- There in:
  - Client: send a request to the server and wait for an answer from the server.
  - Server: accept all valid requests coming from any clients, then execute the request and return the results to the client.

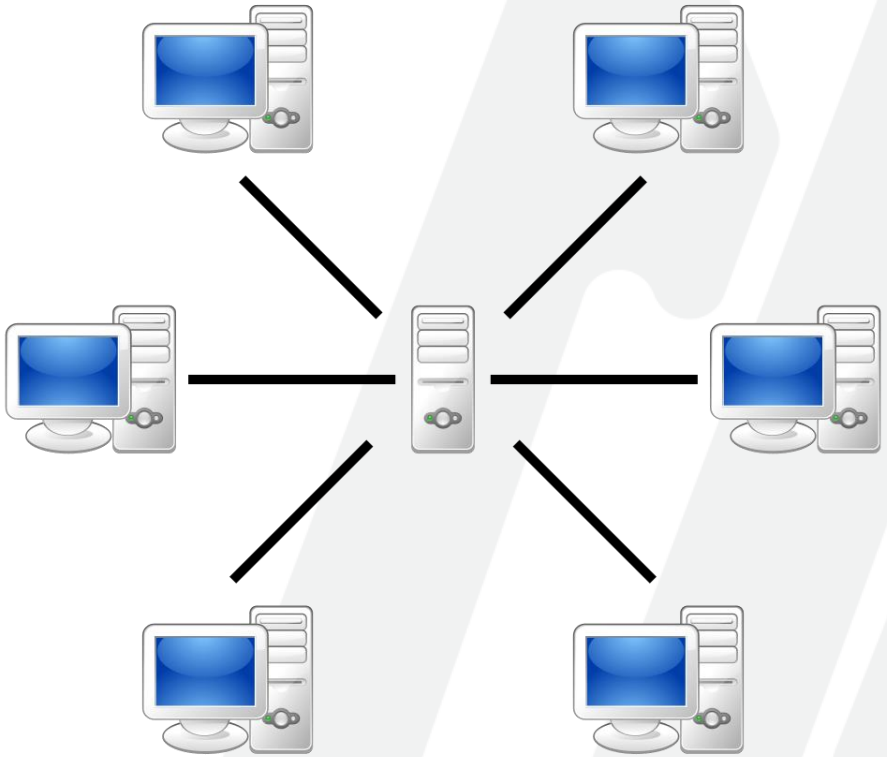
Synchronous vs. Asynchronous

- Synchronous communication:
  - The calling party requests a service and waits for the service to complete.
  - Only when it receives the result of the service it continues with its work.
- Asynchronous communication:
  - The calling party initiates a service call but does not wait for the result.
  - The caller immediately continues with its work without caring for the result.

Synchronous vs. Asynchronous



Model structure



This Photo by Unknown Author is licensed under CC BY-SA

Client-Server architecture

- The application is modeled as **a set of services** that are provided by servers and **a set of clients** that use these services.
- Clients know of servers, but servers need not know of clients.
- Clients and servers are logical processes.
- Examples: email exchange, web access, database access

Client-Server architecture examples

- Clients: web browsers, email clients, and online chat clients.
- Servers: web servers, ftp servers, application servers, database servers, name servers, mail servers, file servers, print servers, and terminal servers.
- Most web services are also types of servers.

The client-server framework

- The client-server framework is also known as a “network computing model”
- All the services and requests are accomplished using the network.
- In the distributed computing systems as all the nodes are doing an independent task. It is a shared computing network where various remotely connected devices send multiple requests to a centralized system (server or host machine).
- The client machine offers a user-friendly design that enables visitors to fire server computer request services and eventually display your performance on the client device.

Client-Side & Server Side

- Client-side
  - An operation performed by the user or clients end in the client-server architecture of a network.
  - Client-side works as a front end like interface on the system;
- Server-side
  - Performs working at the back end.
- Example
  - The web browsers used for requesting pages are considered as client-side
  - The server, from where the requested page will be submitted and will give the data to show on the device will be the server that is hosting the page.

Client-Server Architecture Components

- Workstations
- Servers
- Networking Devices

Workstations

- Workstation is the system of users that are sometimes also named as “client’s computer”.
- These workstations use various kinds of operating systems.
- Although, the OS used on the client’s workstation is much cheaper than the OS considered for server machines.

Server

- An ultra-performer device that retains a fast-processing speed, more storage space and robust memory to deal with multiple requests approaching simultaneously from various workstations.
- At the same time, a server performs numerous kinds of functions, such as mail servers, database servers, file servers, and domain controllers.

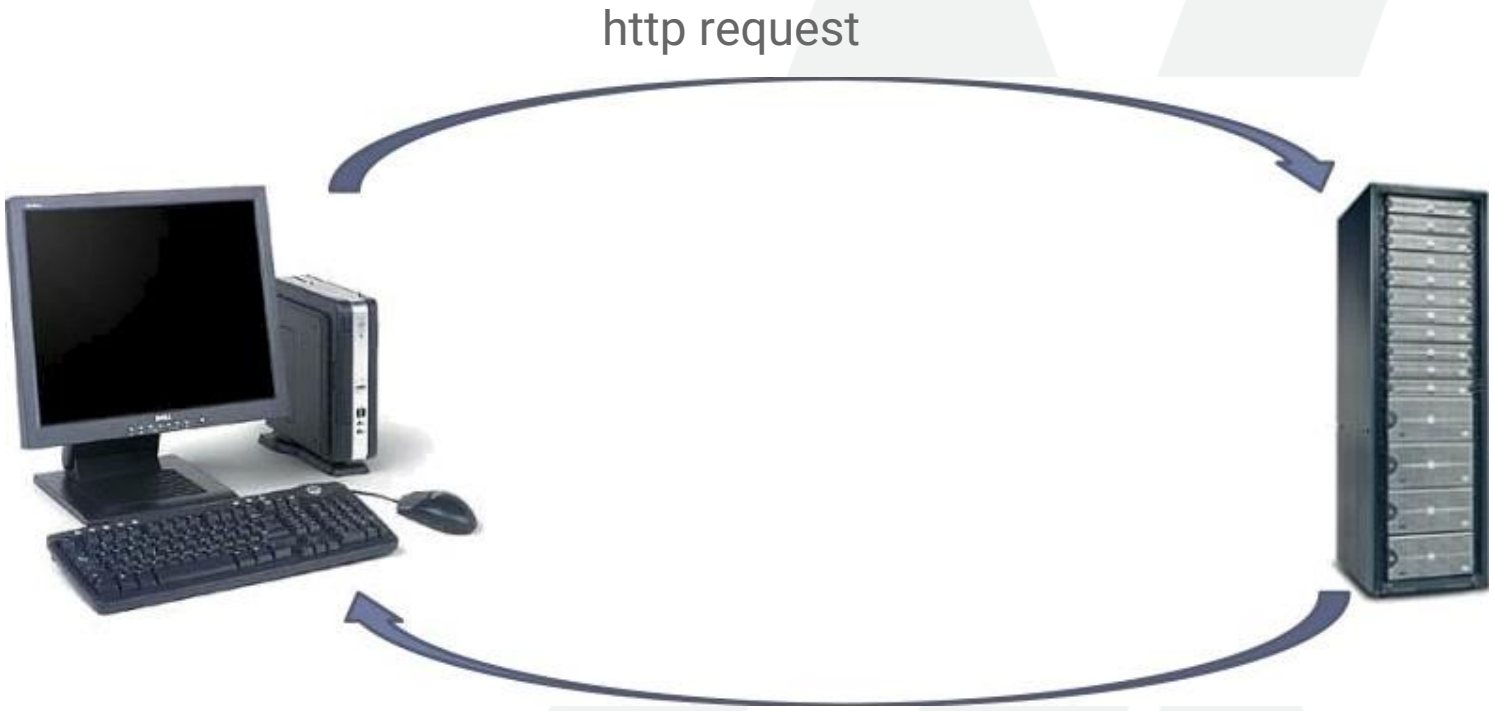
Networking Devices

- Workstations and servers are interconnected with each other by means of a specific medium.
- Such a medium is called a network device.
- Each networking device used in the client-server architecture has its operation and properties.
- Network devices
  - For making a connection to a server with various workstation hubs are used.
  - For transferring data from a device to another device, repeaters are used.
  - For isolated network segmentation bridges are used.

Client-Server Architecture Working

- Clients are considered as users or consumers, whereas the server performs like a producer.
- A client often demands high-end computing services from the server to fulfil the requirements.
- Servers facilitate various services to the client that include access for specific applications, storage of data, sharing of files, utilizing the computing power by means of the server.
- A user at its end sends a request to the server using a network that is processed at various stages and then data is delivered to the user according to the submitted request.
- Servers can entertain multiple consumers simultaneously. Meanwhile, a client must remain connected with the various servers in a single time as each server is providing services according to its domain.

2-Tiers client-server Architecture example



2-Tiers client-server Architecture

- This architecture facility has the best environment.
- It assists in saving the user interface on the system of the user (client) and the related data based are stored on the server device.
- Database logic and business logic needs to be maintained where these logics are to be stored either on the user's end or the server's end.
- When these logics are stored on the client's end, the architecture is called "fat client thin server" however, if these logics are handled at server device, then we call it as "thin client fat server".

Discussion

- What are Thin-client and fat-client models?
- You are giving examples for each model.

Thin and fat clients

- Thin-client model (also called Slim-Client)
  - All the application processing and data management is carried out on the server.
  - The client is simply responsible for running the **presentation software**.
- Fat-client model (Thick-Client)
  - In this model, the server is only responsible for data management.
  - The software on the client implements the **application logic** and the **interactions** with the system user.

Use of Client-Server architectures

- Thin clients
  - Legacy system applications where separating application processing and data management is impractical.
  - Computationally-intensive applications such as compilers with little or no data management.
  - Data-intensive applications (browsing and querying) with little or no application processing.
- Fat clients
  - Applications where application processing is provided by off-the shelf software (e.g., Microsoft Excel) on the client.
  - Applications where computationally-intensive processing of data (e.g., data visualization) is required.
  - Applications with relatively stable end-user functionality used in an environment with well-established system management.

Client-Server Architecture Characteristics

- The architecture operates through a request and response mechanism.
- A request is submitted to the server by the client, and the server provides data in response to requested details.
- A standard contact protocol is adopted by the architecture so that devices can communicate with one another easily. In the application layer, all the data transferring protocols are accessible.
- Only a finite amount of client queries may be accommodated by a server at a particular time. In order to reply to all queries, it employs a method targeting priority.
- In the server-side, denial threats also affect the authenticated request.
- A web server is an example of the architecture programming scheme. It gives the web sites back to the consumers who ordered them.
- As the functionality of both client and server are different; therefore, different hardware and software resources are required for both kinds of machines.

Client-Server Architecture Examples

- Web Servers
- E-Mail Servers
- File Servers
- Domain Name Servers
- Database Servers

Client-Server Architecture Advantages

- There is a centralized network that has full leverage to control the processes and activities in a client-server architecture.
- The central area of the architecture is used for the storage of data.
- The devices used in architecture can be controlled centrally.
- Network protection, data backup and all other concerned elements are tackled centrally.
- Users have the authority to access all the files at any time residing on the central storage.
- There exists no restriction regarding geography to access the information. One can access any information from any place.
- The architecture can be expanded as the growth requires the scaling of users or any other parameters.

Client-Server Architecture Advantages (cont.)

- It offers service incorporation, which ensures that it allows user authorization to view business data through its device and to delete any inappropriate access for logging in.
- Clients have the facility to share any resources at various platforms and places.
- The client-server design is implemented on a distributed model basis, ensuring that the server is replaced, restored, upgraded and moved without influencing the client.
- The architecture can tackle massive weight given clients, systems, devices and network loads.
- The architecture provides an easy user interface, file finding procedure and management system for keeping all the files in an organized pattern.
- The architecture facilitates the data overwhelming by the reduction in data replication.

Client-Server Architecture Disadvantages

- The users will suffer if the primary server goes dump.
- The architecture requires a particular OS related to networking.
- Configuration of hardware components and software tools utilize plenty of costs.
- There is a need for technical as well as skilled staff for maintenance of the network specifically for server machines.
- Numerous requests simultaneously can cause “traffic congestion problems”.
- The architecture can mostly be availed in large organizations and enterprises as the cost is unbearable for typical clients.

Q&A

This Photo by Unknown Author is licensed under CC BY-NC-ND